Design And Analysis of Algorithms
Lecture 19: Johnson-Lindenstrauss

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In this lecture, we consider the following problem. Suppose you are handed a set of vectors $X = \{x_1, \ldots, x_n\} \subset \mathbb{R}^D$, and you would like to reduce the dimension in which you store/do computations on X. Depending upon what you want to retain, this might be possible. In general one need not keep more than $n-1$ dimensions, since n vectors span at most an $n-1$-dimensional subspace, it would suffice to keep around only $n-1$ dimensions to do a variety of computations. We will focus on on maintaining all pairwise distances in X: for this problem, $n-1$ dimensions is also necessary, since there can be n orthogonal vectors that truly do require n dimensions to capture all pairwise distances correctly.

Perhaps you want to capture pairwise distances, and you're willing to have some approximation in measuring them. How many dimensions do we need to represent X such that we approximately preserve pairwise distances? We remind ourselves that pairwise ℓ_2 distances are highly related to the dot product between two vectors, since

$$||x_i - x_j||_2^2 = ||x_i||_2^2 + ||x_j||_2^2 - 2\langle x_i, x_j\rangle$$

and so we will alternate between discussing preserving distances and dot products.

We needed $n-1$ dimensions to exactly store n vectors which were orthogonal (whose dot-products were 0). How many approximately orthogonal (dot products are close to 0) can fit in n dimensions? Suppose $x_{i,t} \sim_{\text{UAR}} \{-\frac{1}{\sqrt{D}}, \frac{1}{\sqrt{D}}\}$ for each $i \in [n], t \in [D]$. Then, $\mathbb{E}[\langle x_i, x_j\rangle] = 0$ and $\max_{i,j}\langle x_i, x_j\rangle \leq 1$, so a simple Chernoff bound tells us that

$$\mathbb{P}[\langle x_i, x_j\rangle \geq \epsilon] \leq e^{\frac{-\epsilon^2}{\sum_n 2/D^2}} = e^{-2\epsilon^2 D}$$

which means that we can take exponentially many vectors and "pack" them into D dimensions, using a union bound. In particular, if $n \leq e^{c\epsilon^2 D}$, if $c < 1$, this term is less than 1. It's also clear that one needs $\Omega\left(\frac{\log n}{\epsilon^2}\right)$ dimensions to maintain all pairwise distances using a basic packing argument.

So, it appears that lots of nearly-orthogonal approximately unit vectors can live in $\mathbb{R}^D$. What about arbitrary sets of vectors, of different lengths and not all (nearly) orthogonal?

1 The Johnson-Lindenstrauss Lemma

The following lemma says it's possible to "flatten" X into many fewer dimensions while maintaining pairwise distances to high fidelity.

Lemma 1 (JL-flattening lemma)**.** *Fix some $D > 1$ and $X \in \mathbb{R}^{D\times n}$. Then, there exists a map $A : \mathbb{R}^D \to \mathbb{R}^k$ with $k = O\left(\frac{\log n}{\epsilon^2}\right)$ such that for all $x_i, x_j \in X$, we have*

$$1 - \epsilon \leq \frac{||A(x_i) - A(x_j)||_2^2}{||x_i - x_j||_2^2} \leq 1 + \epsilon.$$

1.1 The First Construction

Consider drawing a matrix M such that $M_{ij} \sim \mathcal{N}(0,1)$ independently for all $i \in [D], j \in [k]$, and let $A(x) = \frac{1}{\sqrt{k}}Mx$ for all x. We now introduce a distributional version of our original 'goal' lemma, stated below.

Lemma 2 (Distributional JL)**.** *For $\epsilon \in (0, \frac{1}{2})$, and A as above, if $||x||_2^2 = 1$, then for $k \geq \frac{c \log \frac{1}{\delta}}{\epsilon^2}$,*

$$\mathbb{P}[||Ax||_2^2 \in [1 \pm \epsilon]] \geq 1 - \delta.$$

Notice that if we have the above lemma, then by scaling and a union bound over $O(n^2)$ pairs of vectors, we'll have proven Lemma 1. So, how should we show the following? In order to analyze this map, it will be useful to have the following facts about Gaussians.

Definition 1. *The p.d.f. of $\mathcal{N}(\mu, \sigma)$ is*

$$f(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-(x-\mu)^2/2\sigma^2}.$$

Lemma 3. *Given two random variables $y_1 \sim \mathcal{N}(\mu_1, \sigma_1^2), y_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ drawn independently, for any $c \in \mathbb{R}$,*

$$cy_1 \sim \mathcal{N}(c\mu_1, c_1^2\sigma_1^2)$$

and

$$y_1 + y_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

So, we'll want to analyze the distribution of Mx using these facts, since the first part of Lemma 3 says the scaling at the end won't affect our analysis.

Let's start by analyzing $(Mx)_p$, the pth coordinate of this projection for a particular $x \in X$. Then, we have

$$(Mx)_p = \sum_{t=1}^{D} M_{tp} x_t$$

where $M_{tp} \sim \mathcal{N}(0,1)$ independently for each t, and so we know $(Mx)_p \sim \mathcal{N}(0, x_1^2 + \ldots + x_D^2)$ by the second part of Lemma 3. if $||x||_2^2 = 1$, then $(Mx)_p \sim \mathcal{N}(0,1)$, i.e. it looks like a random variable drawn from a standard normal! Moreover, this dimensions of Mx will be independent of the other dimensions, since they use D different (independent) draws from a Gaussian.

1.2 Average length of a unit vector's projection

So, what is the squared length of $Ax = \frac{1}{\sqrt{k}}Mx$? Define

$$Z = \frac{1}{k} \sum_{i=1}^{k} R_i^2$$

where $R_i \sim N(0,1)$ independently. By the previous reduction,

$$\mathbb{E}[||Ax||_2^2] = \mathbb{E}[Z]$$

and since $\mathbb{E}[R_i^2] = 1$, we know $\mathbb{E}[Z] = 1$.

1.3 Concentration about the mean

If we had bounded random variables R_i, we'd be all set: we could apply a Chernoff bound. Instead, they're Gaussian and have infinite support, so we'll redo some of the work we did proving Chernoff. For any $t > 0$, consider the standard exponentiation technique:

$$\mathbb{P}[Z \geq 1+\epsilon] \leq \mathbb{P}[e^{tkZ} \geq e^{tk(1+\epsilon)}] \leq \frac{\mathbb{E}[e^{tkZ}]}{e^{tk(1+\epsilon)}} = \prod \frac{\mathbb{E}[e^{tkR_i^2}]}{e^{t(1+\epsilon)}}.$$

Now, it's time to employ our knowledge of the CDF of the normal distribution to analyze the numerator:

$$\mathbb{E}[e^{tkR_i^2}] = \frac{1}{\sqrt{2\pi}} \int_{x\in\mathbb{R}} e^{tx^2} e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}} \int_{z\in\mathbb{R}} \frac{e^{-z^2/2}}{\sqrt{1-2t}} dz = \frac{1}{\sqrt{1-2t}}$$

and so

$$\mathbb{P}[Z \geq 1+\epsilon] \leq \left(\frac{1}{e^{t(1+\epsilon)}\sqrt{1-2t}}\right)^k.$$

Notice that

$$\begin{aligned}
\frac{1}{e^t\sqrt{1-2t}} &= e^{-t-\frac{1}{2}\log(1-2t)} \\
&= e^{(2t)^2/2+(2t)^3/6+\ldots} \\
&\leq e^{t^2(1+2t+2t^2+\ldots)} \\
&\leq e^{t^2/(1-2t)}
\end{aligned}$$

and putting these both together, we have that

$$\mathbb{P}[Z \geq 1+\epsilon] \leq e^{kt^2/(1-2t)-kt\epsilon} \leq e^{-k\epsilon^2/8}.$$

So for k as in the theorem statement, this is at most $\delta/2$, as desired.

2 What did we need about M?

We needed that the coordinates were drawn independently, with mean 0, and that their tails were not too extreme. In particular, we used that

$$\mathbb{E}[e^{tR_i^2}] \leq \frac{1}{\sqrt{1-2t}}$$

which is related to being a subgaussian distribution.

We won't work this out, but it'd suffice to consider *Rademacher* random variables, which are ± 1 with probability $\frac{1}{2}$, primarily because this distribution is subgaussian. To see more references, see `https://www.cs.cmu.edu/~anupamg/advalgos17/`.

3 Lower bounds

There has been a line of work showing that JL is (nearly optimal): first we learned we neeed at least $\tilde{\Omega}\left(\frac{\log n}{\epsilon^2}\right)$ dimensions [Alon], then that if we were restricted to linear projections that we needed $\Omega\left(\frac{\log n}{\epsilon^2}\right)$ dimensions [Green Larsen and Nelson, 2014], and recently this restriction to linear mappings was removed [Larsen and Nelson, 2017]

4 Compressed Sensing and Basis Pursuit

One nice application of Johnson-Lindenstrauss is in the area of compressed sensing: given a set of *measurements* $Ax = b$ for some sparse $x \in \mathbb{R}^D$, is it possible to (approximately) recover x? Suppose $||x||_0 \leq r$, which we call r-spariety. Is it possible to reconstruct x using only something like r measurements of x? For worst-case A, this problem is NP-hard, but what if we can choose A, knowing we'd like to be able to reconstruct x?

A related problem, called Basis Pursuit (BP), asks the following question. Minimize the ℓ_1 norm of a vector x, subject to $Ax = 1$. This can be written as a linear program, by introducing variables y_i and adding the constraint $-y_i \leq x_i \leq y_i$ for each n and minimizing the sum of the y_is.

So what do BP and/or compressed sensing have to do with Johnson-Lindenstrauss? As it turns out, if you have a distribution over matrices which satisfy Lemma 2, then that distribution will do a good job with both of these problems. To make this precise, we introduce the following definition:

Definition 2 (BP-exact)**.** *A is called BP-exact if for all r-sparse x^*, x^* is the unique solution to basis pursuit for $Ax^* = b$.*

Theorem 1 (Donoho [2006],Candes and Tao [2005])**.** *IF A is drawn from a distributional JL family for $k \geq c \cdot r \log \frac{D}{r}$, then with high probability A is BP-exact.*

To prove the concentration this theorem suggests, one needs to argue about the following definition.

Definition 3. *A matrix A is (t, ϵ)-RIP (restricted isometry property) if for all t-sparse unit vectors x,*

$$||Ax||_2^2 \in [1 \pm \epsilon].$$

References

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